

TrES-1b

R-1742

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_190827 · created 2026-08-02T18:42:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458722.7473 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.0079
Transit depth (Rp/Rs)^2	2.22 +/- 0.23 [%]
Orbital Inclination (inc)	89.84 +/- 0.52
Airmass coefficient 1 (a1)	0.9764 +/- 0.01
Airmass coefficient 2 (a2)	0.0203 +/- 0.0084
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#1 - [627, 225]
Optimal Aperture	4.26
Optimal Annulus	10.61
Transit Duration (day)	0.10512 +/- 0.00096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.0079 vs 0.1358 (deviation 1.67σ)
Duration fit vs published	0.10512 d vs 0.1042 d (deviation 0.99σ)
Mid-time O–C	3.6 min
Depth SNR	18.9
Residual scatter	1.02 %

Light curve

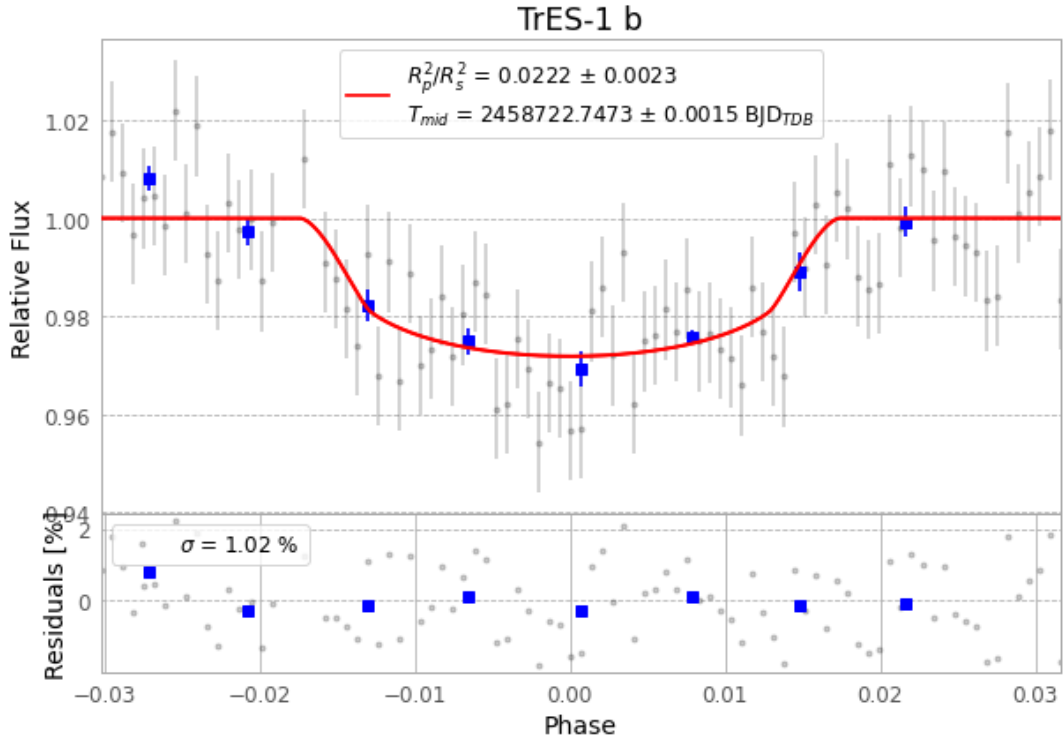


Figure 1 — FinalLightCurve_TrES-1 b_2019-08-26.png (SHA-256 c9a439f52ae5c773...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a868e1751a80af21...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), TRES-1190827033912.FITS → TRES-1190827080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190827.log (8d4ad803575b...), FinalLightCurve_TrES-1 b_2019-08-26.png (c9a439f52ae5...), FinalLightCurve_TrES-1 b_2019-08-26.csv (7c8f602723f4...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:42Z.